Decidable languages related to CFG's

We will consider following languages today.

$A_{CFG} = \{\langle G, w\rangle \mid G \text{ is a CFG and } G \text{ generates } w\}$

$E_{CFG} = \{\langle G\rangle \mid G \text{ is a CFG and } L(G) = \phi\}$

$EQ_{CFG} = \{\langle G, H\rangle \mid G, H \text{ are CFGs and } L(G) = L(H)\}$

$A_{CFG} = \{\langle G, w\rangle \mid G$ is a CFG an w is a string and G generates $w\}$

- go through every possible derivation of grammar G and check whether w can be generated.
- we do not have a bound on the # of derivations
- what if we convert G to CNF form?

Theorem 4.7

A_{CFG} is decidable

- If G were in Chomsky Normal form, any derivation of w where $|w| = n$, has $2n-1$ steps.

(Assignment 04 problem 05)

Proof: construct TM S as follows:

S = "On <G,w> where G is CFG & w is a string:

1. Convert G into CNF.
2. List all derivations with $2n-1$ steps, where $n = |w|$
3. If any of the derivations generates w, accept; otherwise, reject."

is this an efficient al orithm?

No, we may have to look at exponential # of derivations

$E_{CFG} = \{ \langle G \rangle \mid G \text{ is a CFG and } L(G) = \emptyset \}$

- Use A_{CFG} and try all possible $w \in \Sigma^*$
- we try to check whether start variable can generate at least one string

The idea is simple. We check whether start variable can generate a string of terminals.

Theorem 4.8 E_{CFG} is decidable.

construct TM R for E_{CFG}:

R = "On input $\langle G \rangle$

1. Mark all terminals in G.
2. Repeat until no new variables get marked
   - Mark A, if $A \to u_1 u_2 \dots u_k$ is a rule in G and all u_i's are marked.
3. If the start variables is not marked, accept; otherwise, reject."

Ex: $S \to ABX \mid BAX$

$A \to AX$

$B \to BX$

$X \to a$

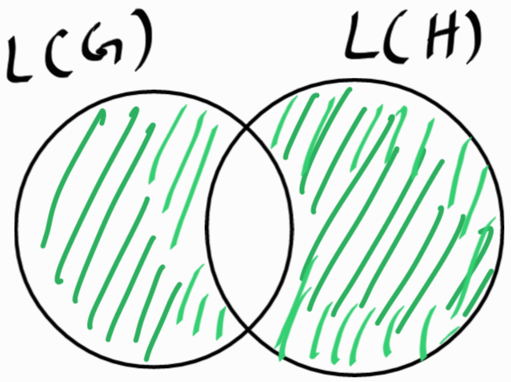
$\underline{X} \to a$, $B \to BX$, $A \to AX$,

$S \to ABX$, $S \to BAX$

$$EQ_{CFG} = \{\langle G, H\rangle \mid G, H \text{ are CFG's and } L(G) = L(H)\}$$

Can we follow the idea used in EQ_{DFA}?

EQ_{CFG}: Given two CFGs, are they equal?



$$L(C) = (L(G) \cap \overline{L(H)}) \cup (L(H) \cap \overline{L(G)})$$

$$L(G) = L(H) \iff L(C) = \emptyset$$

Q: Are CFG's closed under $\cup, \cap, \overline{}$ operators?

Claim 1: CFL's are closed under union operation.

Proof Idea:

G: $S_1 \to \dots$

H: $S_2 \to \dots$

Construct G' s.t

$$S \to S_1 \mid S_2$$

$S_1 \to \dots$

$S_2 \to \dots$

claim 2:

CFL's are <u>not</u> closed under

$\cap$, $-$ operation.

we only need one counter example.

$A = \{a^i b^j c^k \mid i, k, j \geq 0, i = j \text{ or } j = k\}$

* this is a CFL

$B = \{a^m b^m c^n \mid m, n \geq 0\}$ * this is a CFL

$C = \{a^n b^m c^m \mid m, n \geq 0\}$ * this is a CFL

$D = \{a^n b^n c^n \mid n \geq 0\}$ * this is not a CFL

$D = B \cap C$

Claim 02: CFL's are <u>not</u> closed under complement ($\overline{}$) operation.

consider CFGs B and C

Let $L = B \cap C$

$B \cap C = \overline{\overline{B} \cup \overline{C}}$ (De morgan's law)

If $\overline{}$ operator is closed then $B \cap C$ should be a CFL. But this is not the case.

This is a counter example.

$\overline{}$ operator should not be closed for CFL's.

- Later we show that EQ_{CFG} is not decidable.